6-3. Multiple regression: Hypothesis tests and Confidence intervals

Consider now the normal model. That is,

$$\boldsymbol{\epsilon} \quad \sim \quad N\left(\mathbf{0},\, \sigma^2 I_n\right)$$

It can be shown that

$$\frac{\hat{\beta}_i - \beta_i}{s.e.\left(\hat{\beta}_i\right)} \quad \sim \quad t_{n-k-1}$$

where

$$s.e.\left(\hat{\beta}_i\right) \quad = \quad \hat{\sigma}\sqrt{c_{ii}}$$

and c_{ii} is the ith diagonal element of $\left(X^T X\right)^{-1}$

Notation: rows/columns are numbered from zero, so the top left diagonal element is c_{00}

Example 6.2

For the data of example 6.1, test whether the intercept is non-zero.

First, recall that

$$\widehat{\sigma^2} = \frac{1}{n-k-1}\left(\boldsymbol{Y}^T\boldsymbol{Y} - \hat{\boldsymbol{\beta}}^T X^T \boldsymbol{Y}\right)$$

Using R, compute

$$\boldsymbol{Y}^T\boldsymbol{Y} = 67.44, \qquad \hat{\boldsymbol{\beta}}^T X^T \boldsymbol{Y} = 67.1031,$$
$$\widehat{\sigma^2} = \frac{1}{7-2-1}(67.44 - 67.1031) = 0.08422$$

$$\Rightarrow \quad \hat{\sigma} = 0.2902$$

We previously computed $\left(X^T X\right)^{-1}$

```
> XTXinverse
             [,1]        [,2]         [,3]
[1,]  6.68309535 -1.52924919 -0.063749424
[2,] -1.52924919  0.76001842 -0.028558268
[3,] -0.06374942 -0.02855827  0.005315523
```

So that

$$c_{00} = 6.683, \quad c_{11} = 0.760, \quad c_{22} = 0.0053$$

and then

$$s.e.\left(\hat{\beta}_0\right) \quad = \quad \hat{\sigma}\sqrt{c_{00}} \quad = \quad 0.2902 \times \sqrt{6.683} \quad = \quad 0.7502$$

We have (from earlier) $\hat{\beta}_0 = -0.2138$

Want to test

$$\begin{aligned} H_0 &: \beta_0 = 0 \\ \text{versus } H_1 &: \beta_0 \neq 0 \end{aligned}$$

Test statistic is

$$t = \frac{\hat{\beta}_0}{s.e.\left(\hat{\beta}_0\right)} = \frac{-0.2138}{0.7502} = -0.285$$

4 degrees of freedom, so for two-sided test at 10% level tables give critical value $t_4(95\%) = 2.132$. Cannot reject H_0 at 10% level.

No evidence of non-zero intercept.

Or directly from output of `lm` shown earlier.

```
> summary(regression)

Call:
lm(formula = Y ~ x1 + x2)

Residuals:
       1        2        3        4        5        6        7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x1           0.89843    0.25300   3.551   0.0238 *
x2           0.01745    0.02116   0.825   0.4558
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

Hypothesis tests and Confidence intervals

Can go for through same process for slope parameters β_1, β_2

Confidence intervals can be computed in the usual way.

Eg 95% CI for β_1 is

$$\hat{\beta}_1 \pm t_{n-k-1}(2.5\%) \times s.e.\left(\hat{\beta}_1\right)$$
$$= \quad 0.8984 \pm 2.776 \times 0.2530 \quad = \quad [0.196, 1.601]$$

Or in R:

```
> confint(regression, level=0.95)
                  2.5 %     97.5 %
(Intercept) -2.29680563 1.86916860
x1           0.19599296 1.60087484
x2          -0.04129219 0.07619777
```

Variance estimation

Recall

$$\widehat{\sigma^2} = \frac{SS_{RES}}{n-k-1} = \frac{1}{n-k-1}\left(\boldsymbol{Y}^T\boldsymbol{Y} - \hat{\boldsymbol{\beta}}^T X^T \boldsymbol{Y}\right)$$

Similarly to simple linear regression, we have

$$\frac{(n-k-1)\widehat{\sigma^2}}{\sigma^2} \sim \chi^2_{n-k-1}$$

So a 95% CI for σ^2 is

$$\left[\frac{(n-k-1)\widehat{\sigma^2}}{\chi^2_{n-k-1}(97.5\%)}, \frac{(n-k-1)\widehat{\sigma^2}}{\chi^2_{n-k-1}(2.5\%)}\right]$$

Variance estimation

In our example, $\widehat{\sigma^2} = 0.08422$ and $n - k - 1 = 4$

95% CI for σ^2 is

$$\left[\frac{4 \times 0.08422}{\chi^2_4(97.5\%)}, \frac{4 \times 0.08422}{\chi^2_4(2.5\%)}\right]$$

$$= \left[\frac{4 \times 0.08422}{11.14}, \frac{4 \times 0.08422}{0.4844}\right] = [0.030, 0.695]$$

95% CI for σ is

$$[\sqrt{0.030}, \sqrt{0.695}] = [0.174, 0.834]$$

ANOVA

We can also test

$$\begin{aligned} H_0 &: \beta_1 = \beta_2 = \cdots = \beta_k = 0 \\ \text{versus } H_1 &: \beta_i \neq 0 \text{ for at least one } i \in \{1, 2, \ldots, k\} \end{aligned}$$

This tests the overall fit of the model.

Under H_0 (and the normal model) we have

$$F_0 = \frac{SS_{REG}/k}{SS_{RES}/(n-k-1)} \sim F_{k,\, n-k-1}$$

and the p-value for H_0 is given by $\Pr\left(F_{k,\, n-k-1} \geq F_0\right)$

Note that for simple linear regression the hypothesis reduces to $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$, and the test turns out to be the same as the t-test.